

Unit 1 Exam Review Fundamental Concepts

Definitions

Write a definition for each term listed below. Your definition should be clear, concise, and use mathematical vocabulary and nomenclature. You should brainstorm a couple of examples that illustrate the meaning of your definition – but examples alone are not enough.

1. Function (refer to the three example definitions from page 5)
2. Domain, Codomain, Range, and Natural Domain (see page 6)
3. Increasing/Decreasing Function (see page 7)
4. Types of Functions (see page 8)
 - a. Surjective Function
 - b. Injective Function
 - c. Bijective Function
5. Odd and Even Functions (see page 9)
6. Invertible Function & the Inverse of a Function (see page 14)
7. Property of Invertible Functions (see page 16)
8. Sequence (see page 22 and corresponding classwork)
9. Piecewise Function (see page 29)

Important Vocabulary

For the list of terms below, you do not need to provide a definition. Rather, you should understand the meaning of the term and describe this meaning in a few words and how it is related to the concepts you have learned so far.

1. Vertical Line Test
2. Unique vs. Distinct
3. Cartesian Plane
4. Cartesian Product
5. Mathematical Symbols (discussed in class; not all are present in the handouts)
 - a. $\{ \}$
 - b. $\in$
 - c. $\exists$
 - d. $\forall$
 - e. $\therefore$
6. Identical Function
7. Symmetry about the x -axis, y -axis, origin, and the line $y = x$
8. Composition of Functions, e.g., $(f \circ g)(x)$
9. Explain the difference between $f^{-1}(x)$ and $f(x)^{-1}$
10. What is the *opposite* of a number vs. what is the *reciprocal* of a number?
11. Rate of Change
12. Common Ratio vs. Common Difference
13. Triangular Differences
14. Recognize each pattern below and their corresponding parent function
 - a. Sum-Sum
 - b. Sum-Constant Second Difference
 - c. Product-Product
 - d. Sum-Product
 - e. Product-Sum

Concept Questions

All questions marked with a * should be answered in complete sentences using mathematical vocabulary. The rest can be explained in Layman's terms. Any one of these are a possible question on the free response portion of the exam.

- How do you verify that a relation is a function when given:
 - A table or set of ordered pairs
 - An equation
 - A graph
 - A mapping diagram
- *Let S be the set of all schools in Texas and let D be the set of all school districts in Texas.
 - Explain the meaning of $T = \{(s, d) \in S \times D \mid \text{the school } s \text{ is in the district } d\}$
 - Is $T : S \rightarrow D$ a function?
 - Is $T : D \rightarrow S$ a function?
- *Give the natural domain for the following functions and explain *why* the domain is restricted to that interval.
 - $f(x) = \sqrt[n]{x}$ where n is even.
 - $f(x) = \frac{1}{x}$
 - $f(x) = \log x$
- Given a function f on some interval $[a, b]$, how can you DISPROVE if f is increasing (or decreasing) on the given interval if you are only given the equation of the function? Is there a way to prove it?
- *For each parent function below determine if the function is even, odd, or neither AND classify the function as surjective, injective, bijective, or neither. Explain how you know for each!
 - $y = x$
 - $y = x^2$
 - $y = \sqrt{x}$
 - $y = x^3$
 - $y = \sqrt[3]{x}$
 - $y = |x|$
 - $y = 10^x$
 - $y = \log x$
 - $y = \frac{1}{x}$
- Give an example illustrating a function that is ONLY surjective, ONLY injective, bijective, and neither using:
 - A mapping diagram
 - A graph
- (Optional, but good practice): Using Desmos, graph the following trigonometric functions listed below and determine whether each function is even, odd, or neither by observing the image of the graph (this is a bit of a spoiler for what we study later in the year!).
 - $y = \sin x$
 - $y = \cos x$
 - $y = \tan x$
 - $y = \csc x$
 - $y = \sec x$
 - $y = \cot x$

8. *Given an even function, such as $f(x) = x^2$, determine if f is still an even function after the transformation:
 - a. Horizontal Translation
 - b. Vertical Translation
 - c. Vertical/Horizontal Stretching/Compressing
 - d. Horizontal Reflection
 - e. Vertical Reflection
9. *Consider the same transformations from #8 but for an odd function such as $f(x) = x^3$.
10. *Is it possible for a function to be even and odd? Justify your reasoning (#5 from Exploration 1.2.3).
11. Let $f(x) = x^2$ and let $g(x) = \log x$. Without using a graphing tool, find the domain and range of:
 - a. $f(g(x))$
 - b. $g(f(x))$
 - c. $f(f(x))$
 - d. $g(g(x))$ – This one is tricky; think carefully about when $\log x > 0$ and when $\log x \leq 0$.
12. Let f be any arbitrary even function and let g be any odd function.
 - a. Is f invertible?
 - b. Is g invertible?
 - c. If a function is *not* invertible, is there a way to make it so? How?
13. *If $f(g(x)) = g(f(x)) = x$, what conclusion can you make about the functions f and g ? Create two functions that exemplify this property (you may consider restricting the domain on one or both of the functions).
14. Given a bottle with a Height vs. Volume graph that is created as the bottle is consistently filled with a liquid, describe the correlation between the width of the bottle and the steepness of the graph (taken from Quiz 1.3.2-1.5)
15. The sequence $\{a_n\}$ is given by $\{a_n\} = \{1, 15, 53, 127, 249, 431, \dots\}$.
 - a. *What kind of function could be used to describe $\{a_n\}$? Explain how you know!
 - b. (Challenge question) Can you find an explicit formula for $\{a_n\}$? That is, can you write the equation of the function that generates this sequence of numbers?
16. *Verify each statement (#'s 1–5) from Exploration 1.5.2. It is possible this will appear on the free response portion of the exam OR on the Unit 1 Exploration Assessment.
17. Create piecewise function using three distinct sub-functions (meaning varying parent functions) that has a domain of all real numbers less than -3 or greater than or equal to 0 , and a range of all real numbers greater than 2 .
18. The set of integers is defined as $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$. The ceiling of x , denoted $\lceil x \rceil$, is defined to be the smallest integer k such that $k \geq x$.
 - a. Write a piecewise function that describes the function $f(x) = \lceil x \rceil$.
 - b. Is it true that $\lceil x + k \rceil = \lceil x \rceil + k$, for any $k \in \mathbb{R}$? Can you find examples of when it is true? Can you find any examples where it is not true? If the statement is true, give a logical argument as to why.